

Computational Methods for Finance

code 21040

(Applied Numerical Finance - code 20247)

Mock Exam

Instructions. The exam is ON PAPER, lasts 1 hour and has 2 parts. The first part contains 8 multiple-choice questions, **each of which may have more than one correct answer**. In the second part you have an open exercise to be solved on a separate sheet. You will deliver this exam text with Part 1 and your solutions to Part 2.

PART 1 (16 points out of 31)

1. **(2 points)** Let W be a standard Brownian motion, N a Poisson process, $J(t) = \sum_{j=1}^{N(t)} (Y_j - 1)$ with Y_j i.i.d. *positive* random variables with respect to the historical probability $\mathbb{P}$. Assume all processes and random variables are independent of each other. The value of the jump-diffusive risky asset S satisfies

☐ A $\frac{dS(t)}{S(t)} = \left(\mu - \frac{\sigma^2}{2}\right) dt + \sigma dW(t) + dJ(t)$

☐ B $\frac{dS(t)}{S(t^-)} = \mu dt + \sigma dW(t) + dJ(t)$

☐ C $dJ(t) = Y_j$ if jump j arrives at t

☐ D $\frac{dS(t)}{S(t^-)} = \mu dt + \sigma dW(t) - dJ(t)$

2. **(2 points)** In the multivariate Black Scholes model with d non-dividend paying securities, the pricing functional F of a European derivative with final payoff $f(S(T))$, where $S = (S_1, \dots, S_d)$ satisfies

☐ A $F(t, S_1, \dots, S_d) \leq f(S_1, \dots, S_d)$ for all $t \in (0, T)$, for all $S_i > 0$, $i = 1, \dots, d$

☐ B $\frac{\partial F(t; S)}{\partial t} + \sum_{i=1}^d \frac{\partial F(t; S)}{\partial S_i} \cdot S_i r + \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2 F(t; S)}{\partial S_i \partial S_j} \sigma_i^T \sigma_j S_i S_j = F(t, S) r$ for all $t \in (0, T)$,
 $S = (S_1, \dots, S_d)$ and $S_i > 0$, $i = 1, \dots, d$

☐ C $F(T, S_1, \dots, S_d) = f(S_1, \dots, S_d)$ for all $S_i > 0$, $i = 1, \dots, d$

3. **(2 points)** The *replicating strategy* $(\vartheta_0, \vartheta_1, \dots, \vartheta_d)$ of the derivative as in the point above

☐ A is short the derivative F

☐ B self-financing

☐ C is $\vartheta_i(t) = \frac{\partial F}{\partial S_i}(t, S(t))$ for $i = 1, \dots, d$, with $t \in (0, T)$

☐ D always buy-and-hold

4. **(2 points)** Let X be a generic random variable and let $\{X_i\}_{i=1}^n \stackrel{i.i.d.}{\sim} X$. Let $\mu = \mathbb{E}[X]$ and $\sigma^2 = V[X]$. The Monte Carlo estimator of $\mathbb{E}[X]$, $\hat{\mu}_n$, and its standard error, $\sqrt{V[\hat{\mu}_n]}$, are respectively equal to...

- ☐ A $\hat{\mu}_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i$ and $\sqrt{V[\hat{\mu}_n]} = \frac{\sigma^2}{n}$
- ☐ B $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n X_i$ and $\sqrt{V[\hat{\mu}_n]} = \frac{\sigma^2}{n}$
- ☐ C $\hat{\mu}_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n X_i$ and $\sqrt{V[\hat{\mu}_n]} = \frac{\sigma}{\sqrt{n}}$
- ☐ D $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n X_i$ and $\sqrt{V[\hat{\mu}_n]} = \frac{\sigma}{\sqrt{n}}$

5. **(2 points)** Let f be a generic deterministic function and let $Y = f(\mathcal{N}(0, 1))$. Let $\{Z_i\}_{i=1}^n \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$ and set $Y_i = f(Z_i)$ and $\tilde{Y}_i = f(-Z_i)$. Then...

- ☐ A Y_i is independent of $\tilde{Y}_i$ for any $i = 1, \dots, n$
- ☐ B $Y_i \sim \tilde{Y}_i$ for any $i = 1, \dots, n$
- ☐ C Y_i is independent of $\tilde{Y}_j$ for any $i = 1, \dots, n$ and $j \neq i$
- ☐ D $Y_i \sim \tilde{Y}_j$ for any $i = 1, \dots, n$ and $j \neq i$

6. **(2 points)** In the context of the previous question, denote by $\hat{Y}_n^{AV}$ the Antithetic Monte Carlo (MC) estimator of $\mathbb{E}[Y]$. Then...

- ☐ A $\hat{Y}_n^{AV} = \frac{1}{2n} \sum_{i=1}^n (Y_i + \tilde{Y}_i)$
- ☐ B $\hat{Y}_n^{AV}$ is unbiased but not consistent for $\mathbb{E}[Y]$
- ☐ C $V[\hat{Y}_n^{AV}] = \frac{1}{4n} (2V[\tilde{Y}_i] - 2Cov[Y_i, \tilde{Y}_i])$

- ☐ D when the simulation of the pair $(Y_i + \tilde{Y}_i)$ is almost as expensive as the simulation of two Y_i 's, the Antithetic MC estimator displays a smaller sampling variance than the plain MC estimator when $Cov[Y_i, \tilde{Y}_i] < 0$

7. **(2 points)** Let $f(t, x) : [0, T] \times \mathfrak{R}^+ \rightarrow \mathfrak{R}$ be a deterministic function of two variables. Then...

- ☐ A the backward difference estimator of $f_t = \frac{\partial f(t, x)}{\partial t}$ is $\frac{f(t, x) - f(t - \Delta_t, x)}{\Delta_t}$
- ☐ B the forward difference estimator of $f_x = \frac{\partial f(t, x)}{\partial x}$ is $\frac{f(t, x) - f(t, x + \Delta_x)}{\Delta_x}$
- ☐ C the central difference estimator of $f_t = \frac{\partial f(t, x)}{\partial t}$ is $\frac{f(t + \Delta_t, x) - f(t - \Delta_t, x)}{\Delta_t}$
- ☐ D the central difference estimator of $f_{xx} = \frac{\partial^2 f(t, x)}{\partial^2 x}$ is $\frac{f(t, x + \Delta_x) - 2f(t, x) + f(t, x - \Delta_x)}{\Delta_x^2}$

8. **(2 points)** Relying on the notation of the previous question, consider the Black-Scholes model and let $f(t, x)$ represent the time- t no-arbitrage price of a European call option with maturity T and strike price K when the underlying is worth $x = S_t$. Then...

- ☐ A the **explicit** finite differences method for the numerical solution of the PDE solved by f uses the backward difference estimator for f_x and the central one for both f_t and f_{xx}
- ☐ B without any further refinement, the **explicit** finite difference schemes is (very) often numerically unstable
- ☐ C the **implicit** finite differences method for the numerical solution of the PDE solved by f uses the forward difference estimator for f_t and the central one for both f_x and f_{xx}
- ☐ D without any further refinement, the **implicit** finite difference schemes is (very) often numerically unstable

PART 2 (15 points out of 31)

Suppose S_1 and S_2 are driven by

$$\frac{dS_i(t)}{S_i(t)} = \mu_i dt + \sigma_i^T dW^P(t)$$

for $i = 1, 2$ where W^P is a 2-dimensional standard Brownian motion, μ_1 and μ_2 are real numbers, $\sigma_1^T = [v_1, 0]$ whereas $\sigma_2^T = [\rho v_2, \sqrt{1 - \rho^2} v_2]$ for $\rho \in (-1; 1)$, and $v_1, v_2 > 0$. Denote with r the riskless interest rate, and assume there are no dividends.

1. **(7 points)** Consider the European derivative whose final payoff is

$$X(T) = \begin{cases} 1 & \text{if } S_1(T) > S_2(T) \\ -1 & \text{if } S_1(T) \leq S_2(T) \end{cases}$$

Find the risk neutral probability that this final payoff is equal to 1, i.e. find

$$\mathbb{Q}[S_1(T) > S_2(T)].$$

Compute the initial no arbitrage price of the derivative, $S_X(0)$.

Under what conditions is the initial price $S_X(0)$ positive?

2. **(3 points)** When replicating the derivative of point 1, are you taking a long position or a short position in S_1 ? And what about with respect to S_2 ?
3. **(2 points)** If the correlation increases, does the initial price $S_X(0)$ that you computed in Point 2 increase or decrease?
4. **(3 points)** Compute the price of the derivative of point 1 for $t \in (0, T)$, with an arbitrary t .

Solutions

PART 1

1. Let W be a standard Brownian motion, N a Poisson process, $J(t) = \sum_{j=1}^{N(t)} (Y_j - 1)$ with Y_j i.i.d. *positive* random variables with respect to the historical probability $\mathbb{P}$. Assume all processes and random variables are independent of each other. The value of the jump-diffusive risky asset S satisfies

$$\boxed{\text{B}} \quad \frac{dS(t)}{S(t-)} = \mu dt + \sigma dW(t) + dJ(t)$$

2. In the multivariate Black Scholes model with d non-dividend paying securities, the pricing functional F of a European derivative with final payoff $f(S(T))$, where $S = (S_1, \dots, S_d)$ satisfies

$$\boxed{\text{B}} \quad \frac{\partial F(t; S)}{\partial t} + \sum_{i=1}^d \frac{\partial F(t; S)}{\partial S_i} \cdot S_i r + \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2 F(t; S)}{\partial S_i \partial S_j} \sigma_i^T \sigma_j S_i S_j = F(t, S) r \quad \text{for all } t \in (0, T),$$

$$S = (S_1, \dots, S_d) \quad \text{and } S_i > 0, \quad i = 1, \dots, d$$

$$\boxed{\text{C}} \quad F(T, S_1, \dots, S_d) = f(S_1, \dots, S_d) \quad \text{for all } S_i > 0, \quad i = 1, \dots, d$$

3. The *replicating strategy* $(\vartheta_0, \vartheta_1, \dots, \vartheta_d)$ of the derivative as in the point above

$$\boxed{\text{B}} \quad \text{self-financing}$$

$$\boxed{\text{C}} \quad \text{is } \vartheta_i(t) = \frac{\partial F}{\partial S_i}(t, S(t)) \quad \text{for } i = 1, \dots, d$$

$$\text{with } t \in (0, T)$$

4. **(2 points)** Let X be a generic random variable and let $\{X_i\}_{i=1}^n \stackrel{i.i.d.}{\sim} X$. Let $\mu = \mathbb{E}[X]$ and $\sigma^2 = V[X]$. The Monte Carlo estimator of $\mathbb{E}[X]$, $\hat{\mu}_n$, and its standard error, $\sqrt{V[\hat{\mu}_n]}$, are respectively equal to... Denote by $\hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n X_i$ the Monte Carlo (MC) estimator of μ . Then...

$$\boxed{\text{D}} \quad \hat{\mu}_n = \frac{1}{n} \sum_{i=1}^n X_i \quad \text{and} \quad \sqrt{V[\hat{\mu}_n]} = \frac{\sigma}{\sqrt{n}}$$

5. **(2 points)** Let f be a generic deterministic function and let $Y = f(\mathcal{N}(0, 1))$. Let $\{Z_i\}_{i=1}^n \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$ and set $Y_i = f(Z_i)$ and $\tilde{Y}_i = f(-Z_i)$. Then...

$$\boxed{\text{B}} \quad Y_i \sim \tilde{Y}_i \quad \text{for any } i = 1, \dots, n$$

$$\boxed{\text{C}} \quad Y_i \text{ is independent of } \tilde{Y}_j \quad \text{for any } i = 1, \dots, n \text{ and } j \neq i$$

$$\boxed{\text{D}} \quad Y_i \sim \tilde{Y}_j \quad \text{for any } i = 1, \dots, n \text{ and } j \neq i$$

It is well known that, for any i , $Y_i \sim \tilde{Y}_i$ but the two are not independent of each other. However, when $j \neq i$, the independence of Z_i and Z_j implies the independence of $Y_i = f(Z_i)$ and $\tilde{Y}_j =$

$f(-Z_j)$. This property is indeed used in the computation of the sampling variance of the Antithetic MC estimator, when writing $V \left[\sum_{i=1}^n \frac{(Y_i + \tilde{Y}_i)}{2n} \right] = \sum_{i=1}^n V \left[\frac{(Y_i + \tilde{Y}_i)}{2n} \right]$. Additionally, even when $j \neq i$, $Y_i \sim f(Z_i) \sim f(Z_j) \sim f(-Z_j) \sim \tilde{Y}_j$.

6. **(2 points)** In the context of the previous question, denote by $\hat{Y}_n^{AV}$ the Antithetic Monte Carlo (MC) estimator of $\mathbb{E}[Y]$. Then...

A $\hat{Y}_n^{AV} = \frac{1}{2n} \sum_{i=1}^n (Y_i + \tilde{Y}_i)$

D when the simulation of the pair $(Y_i + \tilde{Y}_i)$ is almost as expensive as the simulation of two Y_i 's, the Antithetic MC estimator displays a smaller sampling variance than the plain MC estimator when $Cov[Y_i, \tilde{Y}_i] < 0$

Indeed, $\hat{Y}_n^{AV} = \frac{1}{2n} \sum_{i=1}^n (Y_i + \tilde{Y}_i)$ is both unbiased and consistent for $\mathbb{E}[Y]$ and $V[\hat{Y}_n^{AV}] = \frac{1}{4n} (2V[\tilde{Y}_i] + 2Cov[Y_i, \tilde{Y}_i])$. Moreover, $V[\hat{Y}_n^{AV}] < V[\hat{Y}_{2n}^{MC}]$ (which is the relevant comparison when the simulation of the pair $(Y_i + \tilde{Y}_i)$ is almost as expensive as the simulation of two Y_i 's) if and only if $Cov[Y_i, \tilde{Y}_i] < 0$.

7. Let $f(t, x) : \mathbb{R}^+ \times \mathbb{R} \rightarrow \mathbb{R}$ be a deterministic function of two variables. Then...

A the backward difference estimator of $f_t = \frac{\partial f(t, x)}{\partial t}$ is $\frac{f(t, x) - f(t - \Delta_t, x)}{\Delta_t}$

D the central difference estimator of $f_{xx} = \frac{\partial^2 f(t, x)}{\partial x^2}$ is $\frac{f(t, x + \Delta_x) - 2f(t, x) + f(t, x - \Delta_x)}{\Delta_x^2}$

since the forward difference estimator of $f_x = \frac{\partial f(t, x)}{\partial x}$ is $\frac{f(t, x + \Delta_x) - f(t, x)}{\Delta_x}$ while the central difference estimator of $f_t = \frac{\partial f(t, x)}{\partial t}$ is $\frac{f(t + \Delta_t, x) - f(t - \Delta_t, x)}{2\Delta_t}$.

8. Relying on the notation of the previous question, consider the Black-Scholes model and let $f(t, x)$ represent the time- t no-arbitrage price of a European call option with maturity T and strike price K when the underlying is worth $x = S_t$. Then...

B without any further refinement, the **explicit** finite difference schemes is (very) often numerically unstable

C the **implicit** finite differences method for the numerical solution of the PDE solved by f uses the forward difference estimator for f_t and the central one for both f_x and f_{xx}

indeed, the **explicit** finite differences method for the numerical solution of the PDE solved by f uses the backward difference estimator for f_t (and not for f_x) and the central one for both f_x and f_{xx} . Moreover, the implicit finite difference scheme is numerically stable.

PART 2

1. The risk neutral probability that this final payoff is equal to 1 is

$$\begin{aligned}
\mathbb{Q}[S_1(T) > S_2(T)] &= \mathbb{Q}\left[\ln S_1(0) + \left(r - \frac{v_1^2}{2}\right)T + \sigma_1 W^\mathbb{Q}(T) > \ln S_2(0) + \left(r - \frac{v_2^2}{2}\right)T + \sigma_2 W^\mathbb{Q}(T)\right] \\
&= \mathbb{Q}\left[(\sigma_1 - \sigma_2)W^\mathbb{Q}(T) > \ln \frac{S_2(0)}{S_1(0)} + \left(\frac{v_1^2}{2} - \frac{v_2^2}{2}\right)T\right] \\
&= \mathbb{Q}\left[\|\sigma_1 - \sigma_2\| \sqrt{T}Z > \ln \frac{S_2(0)}{S_1(0)} + \left(\frac{v_1^2}{2} - \frac{v_2^2}{2}\right)T\right]
\end{aligned}$$

as

$$\begin{aligned}
(\sigma_1 - \sigma_2)W^\mathbb{Q}(T) &\sim \|\sigma_1 - \sigma_2\| \sqrt{T}Z \\
\|\sigma_1 - \sigma_2\| &= \sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2}
\end{aligned}$$

with Z standard normal random variable. Thus

$$\begin{aligned}
\mathbb{Q}[S_1(T) > S_2(T)] &= \mathbb{Q}\left[\|\sigma_1 - \sigma_2\| \sqrt{T}Z > \ln \frac{S_2(0)}{S_1(0)} + \left(\frac{v_1^2}{2} - \frac{v_2^2}{2}\right)T\right] \\
&= \mathbb{Q}\left[Z > \left(\ln \frac{S_2(0)}{S_1(0)} + \left(\frac{v_1^2}{2} - \frac{v_2^2}{2}\right)T\right) \frac{1}{\|\sigma_1 - \sigma_2\| \sqrt{T}}\right] \\
&= \mathbb{Q}\left[Z < \left(\ln \frac{S_1(0)}{S_2(0)} - \left(\frac{v_1^2}{2} - \frac{v_2^2}{2}\right)T\right) \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}}\right] \\
&= N\left(\left(\ln \frac{S_1(0)}{S_2(0)} - \left(\frac{v_1^2}{2} - \frac{v_2^2}{2}\right)T\right) \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}}\right) = \alpha
\end{aligned}$$

The initial no-arbitrage price of the derivative is

$$\begin{aligned}
S_X(0) &= e^{-rT} (1 \cdot \mathbb{Q}[S_1(T) > S_2(T)] - 1 \cdot \mathbb{Q}[S_1(T) \leq S_2(T)]) \\
&= e^{-rT} (1 \cdot \mathbb{Q}[S_1(T) > S_2(T)] - (1 - \mathbb{Q}[S_1(T) > S_2(T)])) \\
&= e^{-rT} (2\mathbb{Q}[S_1(T) > S_2(T)] - 1) \\
&= e^{-rT} \left(2N\left(\left(\ln \frac{S_1(0)}{S_2(0)} - \left(\frac{v_1^2}{2} - \frac{v_2^2}{2}\right)T\right) \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}}\right) - 1\right)
\end{aligned}$$

The initial price of the derivative is

$$\begin{aligned}
S_X(0) &= e^{-rT} (2\alpha - 1) \\
\text{with } \alpha &= N\left(\left(\ln \frac{S_1(0)}{S_2(0)} - \left(\frac{v_1^2}{2} - \frac{v_2^2}{2}\right)T\right) \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}}\right)
\end{aligned}$$

The initial price $S_X(0)$ is positive if and only if

$$\begin{aligned}
2\alpha - 1 &> 0 \\
\alpha &> 0.5 \\
\left(\ln \frac{S_1(0)}{S_2(0)} - \left(\frac{v_1^2}{2} - \frac{v_2^2}{2}\right)T\right) &> 0.
\end{aligned}$$

The probability α is increasing in $S_1(0)$ and decreasing in $S_2(0)$. Since

$$\frac{\partial}{\partial \alpha} S_X(0) = 2e^{-rT} > 0,$$

we get that

$$\frac{\partial}{\partial S_1(0)} S_X(0) \geq 0 \text{ and } \frac{\partial}{\partial S_2(0)} S_X(0) \leq 0,$$

i.e. when replicating the derivative of point 1 we are long in S_1 and short in S_2 .

2. The initial price $S_X(0) = e^{-rT} (2\alpha - 1)$ depends on the correlation ρ as

$$\frac{\partial}{\partial \rho} S_X(0) = \frac{\partial}{\partial \alpha} S_X(0) \frac{\partial \alpha}{\partial \rho} = 1e^{-rT} \frac{\partial \alpha}{\partial \rho}.$$

Hence

$$\frac{\partial}{\partial \rho} S_X(0) > 0 \text{ iff } \frac{\partial \alpha}{\partial \rho} > 0$$

Since the term $\frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}}$ is increasing with respect to ρ , the derivative

$$\frac{\partial \alpha}{\partial \rho} = \frac{\partial}{\partial \rho} N \left(\left(\ln \frac{S_1(0)}{S_2(0)} - \left(\frac{v_1^2}{2} - \frac{v_2^2}{2} \right) T \right) \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T}} \right)$$

is positive if and only if

$$\begin{aligned} \left(\ln \frac{S_1(0)}{S_2(0)} - \left(\frac{v_1^2}{2} - \frac{v_2^2}{2} \right) T \right) &> 0 \\ \ln \frac{S_1(0)}{S_2(0)} &> \left(\frac{v_1^2}{2} - \frac{v_2^2}{2} \right) T, \end{aligned}$$

and decreasing otherwise. Hence the sensitivity with respect to the correlation parameter ρ depends on the current prices of the underlying risky securities and their variance rates.

3. The intermediate no-arbitrage price of the derivative is

$$S_X(t) = e^{-r(T-t)} (2\mathbb{Q}_t[S_1(T) > S_2(T)] - 1)$$

We compute

$$\mathbb{Q}_t[S_1(T) > S_2(T)] = \mathbb{Q}_t \left[S_1(t) \frac{S_1(T)}{S_1(t)} > S_2(t) \frac{S_2(T)}{S_2(t)} \right]$$

As $S_1(t)$ and $S_2(t)$ are measurable at t and as $\frac{S_1(T)}{S_1(t)}$ and $\frac{S_2(T)}{S_2(t)}$ are independent of $\mathcal{F}_t$ we obtain

$$\begin{aligned} \mathbb{Q}_t[S_1(T) > S_2(T)] &= \mathbb{Q}_t \left[\ln S_1(t) + \ln \frac{S_1(T)}{S_1(t)} > \ln S_2(t) + \ln \frac{S_2(T)}{S_2(t)} \right] \\ &= \mathbb{Q} \left[(\sigma_1 - \sigma_2) (W^{\mathbb{Q}}(T) - W^{\mathbb{Q}}(t)) > \ln \frac{S_2(t)}{S_1(t)} + \frac{v_1^2 - v_2^2}{2} (T - t) \right] \\ &= \mathbb{Q} \left[Z > \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T-t}} \left(\ln \frac{S_2(t)}{S_1(t)} + \frac{v_1^2 - v_2^2}{2} (T - t) \right) \right] \\ &= \mathbb{Q} \left[Z < \frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T-t}} \left(\ln \frac{S_1(t)}{S_2(t)} - \frac{v_1^2 - v_2^2}{2} (T - t) \right) \right] \\ &= N \left(\frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T-t}} \left(\ln \frac{S_1(t)}{S_2(t)} - \frac{v_1^2 - v_2^2}{2} (T - t) \right) \right) \end{aligned}$$

and hence

$$\begin{aligned}
S_X(t) &= e^{-r(T-t)} (2\mathbb{Q}_t[S_1(T) > S_2(T)] - 1) \\
&= e^{-r(T-t)} \left(2N \left(\frac{1}{\sqrt{v_1^2 - 2\rho v_1 v_2 + v_2^2} \sqrt{T-t}} \left(\ln \frac{S_1(t)}{S_2(t)} - \frac{v_1^2 - v_2^2}{2} (T-t) \right) \right) - 1 \right)
\end{aligned}$$